Reproduction

Mark scheme and worked solutions for 20 questions: 7 multiple choice, 13 structured.

1 · Part 1

Paper 1 & 2 Reproduction [1]

Answer: B

Explanation

A bacterial cell dividing into two genetically identical cells needs only one parent and involves no gametes, so it is asexual reproduction. A sperm cell fusing with an egg cell is fertilisation, and both the fertilised bird egg and fish releasing gametes come from the fusion of male and female gametes, so A, C and D are all sexual reproduction.

2 · Part 1

Paper 1 & 2 Reproduction [1]

Answer: D

Explanation

Fertilisation is the fusion of a male gamete and a female gamete to produce a zygote. An embryo (A) develops later, after the zygote has divided many times, and a fetus (B) is a later stage still. An ovum (C) is the female gamete itself, not the product of fertilisation.

3 · Part 1

Paper 1 & 2 Reproduction [1]

Answer: A

Explanation

Pollen grains are produced in the anthers, the male parts of the flower. The ovary (B) contains the ovules, the stigma (C) receives pollen, and the style (D) joins the stigma to the ovary and is where the pollen tube grows.

4 · Part 1

Paper 1 & 2 Reproduction [1]

Answer: C

Explanation

After fertilisation each ovule develops into a seed. The ovary (A) develops into the fruit that surrounds the seeds, so it is a common confusion. The stigma (B) and anther (D) usually wither after pollination and do not form seeds.

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5 · Part 1 Paper 1 & 2 Reproduction [1]

Answer: D

Explanation

Sperm swim through the uterus into the oviduct, where they meet an egg that has been released from the ovary, so fertilisation normally takes place in the oviduct. Eggs are produced in the ovary (A) but are not fertilised there; the embryo implants and develops in the uterus (B); the cervix (C) is the entrance to the uterus that sperm pass through.

6 · Part 1 Paper 1 & 2 Reproduction [1]

Answer: A

Explanation

Progesterone maintains the uterus lining. A high progesterone level for many weeks with no menstruation is what happens during pregnancy. Progesterone is low before ovulation (B); if the lining were breaking down (C), progesterone would have fallen and menstruation would occur; if the ovaries had stopped producing hormones (D), this would not explain a high progesterone concentration with no menstruation.

7 · Part 1 Paper 2 only Reproduction [1]

Answer: B

Explanation

FSH, from the pituitary gland, stimulates an egg to mature in the ovary and stimulates the ovaries to produce oestrogen. LH (A) causes ovulation. Progesterone (C) is a hormone produced after ovulation that maintains the uterus lining; it does not stimulate oestrogen production. Testosterone (D) is the male hormone produced by the testes.

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8 · Part 1 Paper 1 & 2 Reproduction [6]

Mark scheme

(a) **[2]**

Any two differences, each comparing both types of reproduction:

- asexual reproduction needs only one parent, but sexual reproduction (usually) needs two parents (1)
- asexual reproduction does not involve gametes or fertilisation, but sexual reproduction involves the fusion of a male gamete and a female gamete (1)
- asexual reproduction produces offspring that are genetically identical (clones), but sexual reproduction produces offspring that show genetic variation (1)

ALLOW asexual reproduction is usually faster

ALLOW a single comparison stated for one type only if the comparison is clearly implied, e.g. "only sexual reproduction involves gametes"

(b) **[2]**

- cut a short piece of stem (with some leaves) from the parent geranium plant (1)
- place the cut end in moist soil or compost (OR water) so that it grows roots (and forms a new plant) (1)

ALLOW dip the cut end in rooting powder (OR rooting hormone)

ALLOW cover with a polythene bag to keep the air humid

(c) **[2]**

- advantage: the new plants are genetically identical to the parent, so all of them will produce very sweet fruit (1)

ALLOW new plants are produced more quickly than from seed

- disadvantage: there is no genetic variation, so if a disease (OR change in the environment) affects one plant it is likely to affect them all (1)

ALLOW overcrowding of plants near the parent; competition with the parent

- One mark for an explained advantage and one mark for an explained disadvantage.

Worked solution

In sexual reproduction two parents each produce gametes. When a male and a female gamete fuse at fertilisation, genetic material from both parents is combined, so the offspring are all slightly different. In asexual reproduction there is one parent and no gametes, so the offspring are genetically identical to the parent.

Runners are a natural method of asexual reproduction. Cuttings are an artificial method: the gardener cuts a short piece of stem with a few leaves from the parent plant and puts the cut end into moist compost. Roots grow from the stem and it becomes a new plant identical to the parent.

For the farmer, the big advantage of runners is that every new plant has exactly the same genes as the sweet-fruited parent, so every new plant will also produce sweet fruit. Seeds come from sexual reproduction, so plants grown from seeds might not. The disadvantage is that, because all the plants are

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genetically identical, none of them has any variation that might help it survive a new disease or a change in conditions, so the whole crop could be lost.

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9 · Part 1 Paper 1 & 2 Reproduction [6]

Mark scheme

(a) **[1]**

- flower P (1)

(b) **[4]**

- anthers hang outside the flower (on long filaments) (1)
- so pollen is exposed to the wind and is easily blown away (OR shaken out) (1)
- stigma is large and feathery (1)
- so it has a large surface area to catch (OR trap) pollen grains carried in the air (1)

> **ALLOW** stigma hangs outside the flower, so it is exposed to air currents carrying pollen, as an alternative to either stigma mark

- Maximum 2 marks for the anthers and maximum 2 marks for the stigma.

(c) **[1]**

- pollen is carried randomly by the wind, so most of it does not reach a stigma (of the same species); large numbers increase the chance that some pollen lands on a stigma (1)

> **ALLOW** most pollen is wasted

Worked solution

Flower P has no bright petals, scent or nectar, so it does not attract insects. Its pollen is small, light and smooth, which is suited to being carried by the wind, so P is wind-pollinated.

Each feature of P helps wind pollination. Anthers that hang outside the flower on long filaments are exposed to the wind, so the wind can shake and blow the pollen away. A large, feathery stigma that hangs outside the flower has a large surface area that acts like a net, catching pollen grains from the moving air.

Wind does not carry pollen to a particular flower. Most pollen grains land on the ground or on other plants and are wasted, so the plant must make huge numbers of pollen grains to make it likely that a few land on a stigma of the same species.

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10 · Part 1 Paper 1 & 2 Reproduction [7]

Mark scheme

(a) **[2]**

- the pollen grains stick to (OR catch on) the hairs or body of the insect (1)
- and are then rubbed off onto (OR stick to) the stigma of the next flower the insect visits (1)

ALLOW pollen is carried from one flower to another by the insect for the first mark

(b) **[3]**

- a pollen tube grows out of the pollen grain (1)
- the pollen tube grows down through the style to the ovary (OR to the ovule) (1)
- the male nucleus (OR male gamete) passes down the pollen tube and fuses with the female nucleus (OR female gamete or egg cell) in the ovule (1)

REJECT the pollen grain moves down the style

(c) **[1]**

- a fruit (1)

(d) **[1]**

- each seed forms from the fusion of different gametes (OR is produced by sexual reproduction), so genetic material from two parents is mixed differently in each seed (1)

ALLOW genetic material comes from two parents

Worked solution

Insects pollinate flowers by accident. Sticky or spiky pollen grains cling to the hairy body of the bee while it feeds. When the bee lands on the next flower of the same species, some of these grains rub off onto that flower's sticky stigma.

Pollination is only the transfer of pollen. Fertilisation comes next: a pollen tube grows out of the pollen grain on the stigma, down through the style and into the ovary, and enters an ovule. The male nucleus travels down the tube and fuses with the female nucleus in the ovule. After fertilisation the ovule develops into a seed and the ovary develops into a fruit around the seeds.

This is sexual reproduction. Each seed is formed from a different pair of gametes, so each seed has a different mix of genetic material from the two parents and the seeds are genetically different from each other.

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11 · Part 1 Paper 1 & 2 Reproduction [7]

Mark scheme

(a) **[2]**

- 19 ÷ 20 × 100 (1)
- 95 % (1)

ALLOW 95 with no unit for full marks only if "%" is clearly implied by the working

(b) **[1]**

- tubes 1 and 2 (1)

(c) **[3]**

- boiling removes the oxygen dissolved in the water (1)
- the layer of oil stops oxygen from the air dissolving in the water (OR reaching the seeds) (1)
- seeds need oxygen for (aerobic) respiration to release energy for germination (OR growth) (1)

REJECT answers that say the seeds were killed by boiling (the seeds were not boiled)

(d) **[1]**

- repeat the experiment (with more tubes for each condition) and calculate a mean (1)

ALLOW use more seeds in each tube

Worked solution

(a) Percentage germinated = (19 ÷ 20) × 100 = 0.95 × 100 = 95 %. Check: 95 % of 20 seeds is 0.95 × 20 = 19 seeds.

(b) To test one condition you change only that condition. Tubes 1 and 2 are both at 20 °C in the dark, and the only difference is that tube 1 is dry and tube 2 is moist. No seeds germinated without water but 19 germinated with water, so water is needed.

(c) Tube 4 has water and is at 20 °C, so the missing condition must be oxygen. Boiling drives out the oxygen dissolved in the water, and the oil layer stops oxygen from the air dissolving back in. Germinating seeds need oxygen for aerobic respiration, which releases the energy needed for the seed to grow, so none germinated. The seeds themselves were not boiled; only the water was, and it was cooled first.

(d) One tube of 20 seeds could give an unusual result by chance. Repeating each condition several times and calculating a mean makes the results more reliable.

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12 · Part 1 Paper 1 & 2 Reproduction [7]

Mark scheme

(a) **[1]**

- the number (OR percentage) of seeds that germinate in a fixed time (1)

ALLOW time taken for seeds to germinate

ALLOW length of root (OR shoot) after a fixed time

(b) **[6]**

Indicative content, one mark each, maximum 6:

- place cotton wool in each Petri dish (1)
- add the same volume of water to each dish, measured with the measuring cylinder (OR keep the cotton wool equally moist) (1)
- put the same number of seeds in each dish, e.g. 20 (1)
- use seeds of the same size (OR age or batch) (1)
- set incubators to a range of at least five different temperatures, e.g. 5, 10, 15, 20, 25 and 30 °C (1)
- keep other conditions the same, e.g. all in the dark (OR same light) (1)
- leave for the same length of time, e.g. 5 days (1)
- count the number of seeds that have germinated (e.g. radicle or root visible) (1)
- repeat at each temperature (OR set up several dishes at each temperature) and calculate a mean (1)
- calculate the percentage germination at each temperature (1)

ALLOW check the dishes daily and add water if needed so they do not dry out

Worked solution

The independent variable is temperature, so it is the only thing Lena should change. The dependent variable is what she measures to see the effect, for example the number of seeds that have germinated after 5 days.

A good plan: put a layer of cotton wool in each Petri dish and add the same volume of water to each, for example 10 cm^3, measured with the measuring cylinder. Place 20 mung bean seeds of similar size from the same packet in each dish. Put the dishes into incubators set at 5, 10, 15, 20, 25 and 30 °C. Keep all the dishes in the dark and check them each day so that the cotton wool stays moist. After 5 days, count how many seeds have germinated, meaning a root has started to grow out of the seed. Set up three dishes at each temperature, then calculate the mean number germinated and the percentage germination at each temperature.

Controlling the water, number of seeds, type of seed, light and time makes the test fair, so any difference in germination is caused by temperature. Using a wide range of temperatures and repeats makes the conclusion more reliable.

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13 · Part 1 Paper 1 & 2 Reproduction [9]

Mark scheme

(a) **[2]**

- the dry mass decreases from day 0 to day 6, from 0.80 g to 0.64 g (1)
- the dry mass then increases from day 6 to day 12, to 0.86 g (1)

ALLOW the lowest dry mass is on day 6 for the first mark

- The first mark needs the decrease to day 6; the second needs the increase after day 6. Data quotes are not required for the marks.

(b) **[2]**

- (0.80 − 0.64) ÷ 0.80 × 100 OR 0.16 ÷ 0.80 × 100 (1)
- 20 % (1)

ALLOW −20 %

(c) **[3]**

- the seedling uses its stored food reserves (OR food stored in the seed, e.g. starch) (1)
- the stored food is broken down (by enzymes) and used in respiration (1)
- respiration releases energy for growth, and carbon dioxide is released (OR mass is lost as carbon dioxide) (1)
- the seedling has no leaves (until day 6), so it cannot photosynthesise to make new food (1)
- Maximum 3 marks.

(d) **[2]**

- the leaves (are above the soil and) carry out photosynthesis (1)
- glucose (OR food) is made faster than it is used in respiration, so new biomass is added (OR used for growth) (1)

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Worked solution

(a) Read the table in order. The dry mass falls from 0.80 g at day 0 to its lowest value, 0.64 g, at day 6. After day 6 it rises again, reaching 0.86 g at day 12, which is higher than at the start.

(b) Change in mass = 0.80 − 0.64 = 0.16 g. Percentage change = (0.16 ÷ 0.80) × 100 = 20 %. Check: 20 % of 0.80 g is 0.16 g, and 0.80 − 0.16 = 0.64 g.

(c) Dry mass is the mass of the seedling without its water, so it measures the organic substances. Until day 6 the seedling has no green leaves above the soil, so it cannot photosynthesise. It survives on the food reserves stored in the seed, such as starch. Enzymes break down these reserves, for example starch into glucose, and the glucose is used in respiration to release energy for growth. Respiration releases carbon dioxide, which leaves the seedling, so the dry mass goes down.

(d) Once the leaves appear, the seedling can photosynthesise. It now makes more glucose than it uses in respiration, and the extra is used to make new cells and tissues, so the dry mass increases.

14 · Part 1 Paper 1 & 2 Reproduction [6]

Mark scheme

(a) **[4]**

- (i) produces sperm (1)

ALLOW produces sperm and testosterone

- (ii) sperm duct (1)
- (iii) produces (OR adds) fluid in which sperm can swim (OR fluid that provides nutrients for sperm) (1)

ALLOW produces semen

- (iv) vagina (1)

(b) **[1]**

- the muscle contracts to push the baby out during birth (1)

(c) **[1]**

- sperm are produced best at a temperature lower than normal body temperature (1)

ALLOW keeps the testes cooler than the body so sperm production is efficient

Worked solution

The testes produce sperm (and the hormone testosterone). The sperm duct carries sperm from the testis to the urethra. The prostate gland adds a fluid to the sperm; this fluid gives the sperm nutrients and a liquid to swim in, and the mixture is called semen. During sexual intercourse, sperm are deposited in the vagina. From there they swim through the cervix and uterus to the oviduct.

The uterus has a thick muscular wall. During pregnancy the embryo grows inside it, and at birth the muscle contracts strongly to push the baby out through the cervix and vagina.

The testes hang outside the main body cavity in the scrotum because sperm production works best at a temperature slightly lower than normal body temperature.

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15 · Part 1 Paper 1 & 2 Reproduction [7]

Mark scheme

(a) **[1]**

- day 12 (1)

(b) **[2]**

- (45 − 12) ÷ 12 × 100 OR 33 ÷ 12 × 100 (1)
- 275 % (1)

ALLOW 280 %

(c) **[2]**

- oestrogen causes the uterus lining to repair (OR rebuild) after menstruation (1)
- oestrogen causes the uterus lining to thicken (OR build up, with blood vessels), ready for implantation of an embryo (1)

ALLOW oestrogen inhibits FSH OR stimulates LH release (Paper 2 content) as an alternative to one mark

(d) **[2]**

- progesterone concentration falls (from 18 arbitrary units on day 21 to 3 arbitrary units on day 28) (1)
- progesterone maintains the uterus lining, so when it falls the lining breaks down (and is lost, which is menstruation) (1)

ALLOW the woman is not pregnant, so progesterone falls, for the first mark

Worked solution

(a) The largest oestrogen value in the table is 45 arbitrary units, on day 12.

(b) Change = 45 − 12 = 33 arbitrary units. Percentage change = (33 ÷ 12) × 100 = 2.75 × 100 = 275 %. Check: 12 + 275 % of 12 = 12 + 33 = 45. A percentage increase can be more than 100 % when the value more than doubles.

(c) After menstruation, oestrogen rises. Oestrogen causes the uterus lining to repair and thicken, building up a lining with plenty of blood vessels, ready to receive an embryo if fertilisation happens.

(d) After ovulation, progesterone rises and keeps the thick lining in place. In this cycle the woman did not become pregnant, so progesterone fell from 18 arbitrary units on day 21 to 3 arbitrary units by day 28. Without progesterone the lining is no longer maintained, so it breaks down and is lost through the vagina. This is menstruation, and it marks day 1 of the next cycle.

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16 · Part 1 Paper 2 only Reproduction [7]

Mark scheme

(a) **[2]**

- FSH stimulates an egg to mature (OR develop) in the ovary (OR in a follicle) (1)
- so a mature egg is available to be released (and fertilised) (1)

> **ALLOW** FSH stimulates the ovaries to produce oestrogen, which builds up the uterus lining, for the second mark

(b) **[2]**

- (18 − 4) ÷ (9 − 1) OR 14 ÷ 8 (1)
- 1.75 mm per day (1)

> **ALLOW** 1.8 mm per day

> **ECF** correct division using an incorrect change in diameter or incorrect time, e.g. 14 ÷ 9 = 1.6 mm per day scores 1 mark

(c) **[2]**

- LH causes ovulation (OR the release of the egg from the ovary or follicle) (1)
- at 18 mm the egg is mature (OR fully developed), so it is released when it is ready to be fertilised (1)

(d) **[1]**

- she may become pregnant with several babies (twins or triplets), which is a risk to her health (OR the babies' health) (1)

> **ALLOW** a multiple pregnancy

Worked solution

(a) FSH is released by the pituitary gland. Its role in the menstrual cycle is to stimulate an egg to mature inside a follicle in the ovary. Amara makes very little FSH, so no egg matures and none can be released. The injections replace the missing FSH so that an egg can develop and later be fertilised.

(b) Change in diameter = 18 − 4 = 14 mm. Time taken = 9 − 1 = 8 days (not 9 days: the time is the difference between the two days). Rate = 14 ÷ 8 = 1.75 mm per day. Check: 4 mm + (1.75 mm per day × 8 days) = 4 + 14 = 18 mm.

(c) LH causes ovulation, the release of the egg from the follicle and ovary. The doctor waits until the follicle is about 18 mm across because by then the egg inside is mature. Giving LH at this point makes the mature egg be released at a known time, ready to be fertilised.

(d) If several eggs mature, several may be released and fertilised, giving a multiple pregnancy such as twins or triplets. This carries greater risks for the mother and the babies.

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17 · Part 1 Paper 2 only Reproduction [8]

Mark scheme

(a) **[2]**

- day 14 (1)
- LH concentration is highest on day 14 (OR there is a surge in LH), and LH causes ovulation (1)
- The reason mark can be scored with an incorrect day only if the reason refers to the LH peak.

(b) **[4]**

Indicative content, one mark each, maximum 4:

- FSH stimulates an egg to mature (OR develop) in the ovary (1)
- FSH stimulates the ovaries to produce oestrogen, so oestrogen rises (from 9 to 52 arbitrary units by day 12) (1)
- rising oestrogen inhibits FSH, so FSH falls (from 10 to 7 arbitrary units between days 5 and 9) (1)
- high oestrogen stimulates the release of LH (1)
- LH rises sharply (OR surges) to 60 arbitrary units on day 14 (1)
- LH causes ovulation, the release of the mature egg (1)

ALLOW the pituitary gland releases FSH and LH

(c) **[2]**

- 60 ÷ 6 (1)
- 10 (times) (1)

Worked solution

(a) Ovulation is triggered by a surge of LH. The LH concentration peaks at 60 arbitrary units on day 14, so ovulation most likely took place on day 14.

(b) The four hormones control one another. At the start of the cycle the pituitary gland releases FSH, which stimulates an egg to mature in the ovary and stimulates the ovary to produce oestrogen. Oestrogen rises from 9 to 52 arbitrary units by day 12. Rising oestrogen has two effects: it inhibits FSH, which is why FSH falls from 10 to 7 arbitrary units between days 5 and 9, and when it is high it stimulates the release of LH. LH therefore rises from 6 on day 9 to 60 on day 14. This LH surge causes ovulation, so the mature egg is released.

(c) LH on day 14 ÷ LH on day 9 = 60 ÷ 6 = 10, so the LH concentration was 10 times greater. Check: 6 × 10 = 60.

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18 · Part 1 Paper 2 only Reproduction [6]

Mark scheme

(a) **[3]**

- the high concentrations of oestrogen and progesterone inhibit the release of FSH (from the pituitary gland) (1)
- so no egg matures (in the ovary) (1)
- LH release (OR the LH surge) is also inhibited, so ovulation does not take place (OR no egg is released) (1)
- so there is no egg to be fertilised (1)
- Maximum 3 marks.

(b) **[2]**

- if a pill is taken late (OR missed), the concentration of the hormones in the blood falls (1)
- FSH (OR LH) is no longer inhibited, so an egg may mature (OR ovulation may occur) and she could become pregnant (1)

(c) **[1]**

- many women forget (OR miss) pills or take them late (1)

ALLOW vomiting or illness stops the pill being absorbed

Worked solution

(a) In a natural cycle, FSH makes an egg mature and an LH surge causes ovulation. Oestrogen and progesterone normally inhibit FSH and LH at certain times of the cycle. The pill keeps both hormones high all the time, so FSH stays inhibited and no egg matures, and there is no LH surge, so no egg is released. With no egg, fertilisation cannot happen.

(b) The pill only works while the hormone concentrations in the blood stay high. If a pill is taken late or forgotten, the concentrations fall. FSH is then no longer inhibited, so an egg could mature and be released, and the woman could become pregnant. Taking the pill at the same time each day keeps the hormone levels steady.

(c) With perfect use only 0.3 in every 100 women become pregnant in a year, but with typical use 9 in every 100 do. The pill itself has not changed, so the difference must come from how it is used: in real life many women sometimes forget a pill or take it late.

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19 · Part 1 Paper 1 & 2 Reproduction [7]

Mark scheme

(a) **[2]**

- the fertilised egg cell (zygote) divides (by mitosis OR cell division) (1)
- repeatedly, to form a ball of many cells that grows (and whose cells become specialised) to form the embryo (1)

ALLOW zygote for fertilised egg cell

(b) **[3]**

- nutrients such as glucose, amino acids (OR fatty acids, vitamins, mineral ions) pass to the embryo (1)
- they diffuse (OR pass) from the mother's blood into the embryo's blood in the placenta (1)
- the mother's blood and the embryo's blood are very close but do not mix (1)
- the nutrients are carried to the embryo in the umbilical cord (1)
- Maximum 3 marks.

REJECT the mother's blood flows into the embryo

(c) **[2]**

- amniotic fluid acts as a shock absorber (OR cushions the embryo) (1)
- protecting the embryo from knocks (OR bumps or mechanical damage) (1)

ALLOW supports the embryo; allows the embryo to move; helps keep the temperature of the embryo stable

Worked solution

(a) The fertilised egg cell is a single cell called a zygote. It divides again and again by cell division to form a ball of cells. This ball implants in the uterus lining, and its cells keep dividing and become specialised to form the tissues and organs of the embryo.

(b) The embryo cannot feed itself, so it gets its food from the mother through the placenta. In the placenta the mother's blood flows very close to the embryo's blood, but the two blood supplies do not mix. Dissolved nutrients such as glucose and amino acids diffuse across from the mother's blood into the embryo's blood. The umbilical cord then carries these nutrients to the embryo.

(c) The embryo floats in amniotic fluid inside a membrane. Liquid is difficult to compress, so the fluid acts like a cushion or shock absorber: if the mother's abdomen is bumped, the force is spread out and the embryo is protected from damage.

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20 · Part 1 Paper 1 & 2 Reproduction [7]

Mark scheme

(a) **[1]**

- testosterone (1)

(b) **[2]**

Any two, one mark each:

- growth of facial hair (OR body hair, pubic hair or underarm hair) (1)
- voice becomes deeper (OR voice breaks) (1)
- muscles develop (OR body becomes more muscular) (1)
- penis and testes grow larger (1)
- testes start to produce sperm (1)

ALLOW shoulders broaden

(c) **[2]**

Any two, one mark each:

- breasts develop (1)
- hips widen (1)
- growth of pubic hair (OR underarm hair) (1)
- menstruation (OR the menstrual cycle) begins (1)

ALLOW more fat is laid down under the skin

(d) **[2]**

Any two, one mark each:

- increase in muscle size (OR strength) (1)
- growth of facial (OR body) hair (1)
- voice becomes deeper (1)
- menstrual cycle becomes irregular (OR stops) (1)

ALLOW breasts become smaller

Worked solution

At puberty the testes start to produce more testosterone, which causes the male secondary sexual characteristics: hair grows on the face and body, the voice deepens, the body becomes more muscular, the penis and testes grow and sperm production begins.

In girls the ovaries produce more oestrogen, which causes the breasts to develop, the hips to widen, pubic and underarm hair to grow and the menstrual cycle to begin.

The drugs in (d) act like testosterone. A female athlete taking them could develop some male characteristics, such as larger and stronger muscles (the reason the drugs are misused), more facial and

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body hair and a deeper voice. They can also disturb her own hormone balance, so her menstrual cycle may become irregular or stop.

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